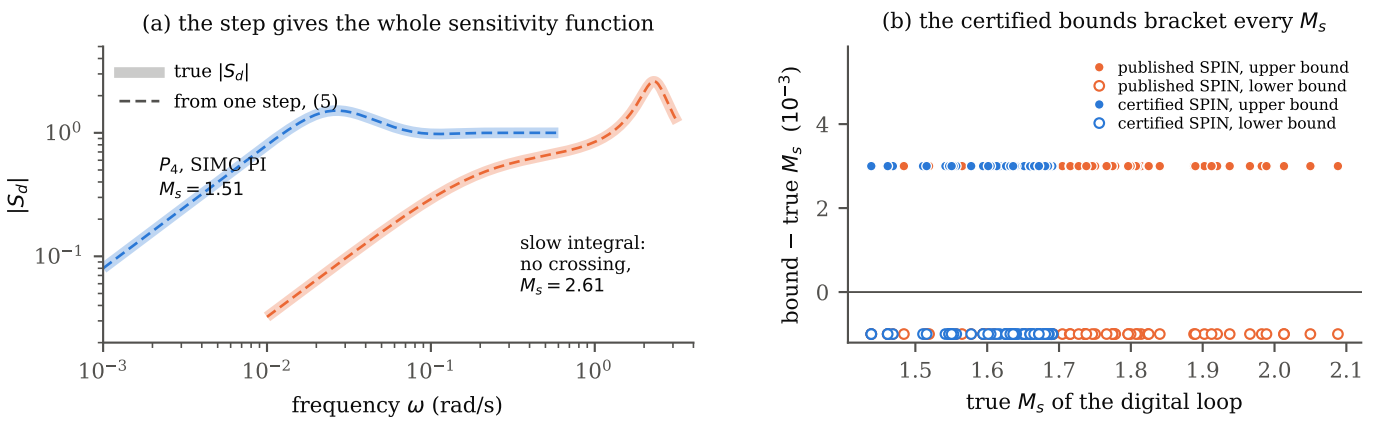
(a) the step gives the whole sensitivity function
true $|S_d|$
from one step, (5)
P_4, SIMC PI
$M_s = 1.51$
slow integral:
no crossing,
$M_s = 2.61$
$|S_d|$
10^0
10^{-1}
10^{-3}
10^{-2}
10^{-1}
10^0
frequency ω (rad/s)
(b) the certified bounds bracket every M_s
published SPIN, upper bound
published SPIN, lower bound
certified SPIN, upper bound
certified SPIN, lower bound
bound − true M_s (10^{-3})
4
2
0
1.5
1.6
1.7
1.8
1.9
2.0
2.1
true M_s of the digital loop